

Measures of Dispersion

The measures of central tendency gives us one single value that represent the entire data but the averages alone cannot describe a set of observations, unless all the observations are the same. For two distributions the central value may be the same but still there can be wide disparities in the formation of the distribution. "The degree to which numerical data tend to spread about an average value is called the variation or dispersion of the data"

Mean is more powerful when it is examined in the light of dispersion. Consider the following data:

Series A	Series B	Series C
100	100	1
100	105	489
100	102	2
100	103	3
100	90	5
500	500	500
$\bar{x} = 100$	$\bar{x} = 100$	$\bar{x} = 100$

In series A, none of the item deviates from mean hence there is no dispersion.

In series B, only one item is perfectly represented by the arithmetic mean, the other vary but variation is small, as compared to C.

In series C, not a single item is represented by arithmetic mean and they are widely spread from mean. In series C, variation or dispersion is much greater compared to B.

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Similarly, if two group of labourer have same mean salary, then equal mean salary is no so important as to know how the salaries are varying from the mean.

Objectives

Objectives of variation:-

- (1) To determine the reliability of an average by pointing out how far an average is representative of entire data.
- (2) Determine the cause of variation and helps to control the variation.
- (3) Helps to compare two or more distributions with regard to their variability.
- (4) Measure of dispersion also facilitates the other ~~set~~ statistical analysis.

Methods of Studying Dispersion :-

The following are the important methods of Studying dispersion:

Absolute measures:-

(1) Range

(2) Quartile Deviation

(3) Mean deviation or average deviation

(4) Variance

(5) standard deviation or Root mean square deviation

Measure of dispersion may be either absolute or relative. The above mention measures of dispersion are absolute measures as they are expressed in same statistical unit in which origin data is given. These measure may be used to compare the two distribution provided the variables are expressed in the same unit and are of same average size.

A relative measure of dispersion is the ratio of the absolute measure of dispersion to an appropriate average. It is called coefficient of dispersion. It has no unit.

While computing the relative dispersion, the average used as base should be same one from which the absolute deviation are measured. i.e. with standard deviation, arithmetic mean should be used. With Mean deviation, arithmetic mean or Median should be used.

The following are relative measure of dispersion :-

- (1) Relative range
- (2) coefficient of quartile deviation
- (3) coefficient of Mean deviation
- (4) coefficient of variation

Range :-

It is simplest method of studying dispersion.

It is defined as difference between the largest and smallest values.

$$\text{Range} = L - S$$

L = largest item

S = Smallest item

For frequency distributed data / the / range is determining / range

For frequency distributed data, the range can be determined using two methods.

Ist Method :- Difference between the upper limit of last class and lower limit of first class

IInd Method :-

Difference b/w the mid point of the highest class and mid point of the lowest class.

The relative measure corresponding to the range is called coefficient of Range, given by.

Coefficient of Range:-

$$\text{coefficient of Range} = \frac{L - S}{L + S}$$

Q1- The prices of shares A B Co. Ltd are given from Monday to Saturday. Find Range and coefficient of Range.

Soln:-

Range:-

Monday	Tues.	Wed.	Thu	Fri.	Sat.
200	210	208	160	220	250

calculate range and its coefficient.

Soln:-

$$\begin{aligned} \text{Range} &= L - S \\ &= 250 - 160 \end{aligned}$$

$$\boxed{\text{Range} = 90}$$

$$\text{coefficient of Range} = \frac{L - S}{L + S}$$

$$= \frac{250 - 160}{250 + 160} = \frac{90}{410}$$

$$\boxed{\text{Coeff. of Range} = 0.22}$$

Q2 Calculate coefficient of Range

Marks	10-20	20-30	30-40	40-50	50-60
No. of Students	8	10	12	8	4

$$\text{here } L = 60 \quad S = 10$$

$$\text{coefficient of Range} = \frac{L - S}{L + S}$$

$$= \frac{50}{70} = 0.714$$

If the two distribution has same average, ~~or~~ or nearly, then comparison of the range indicates that the distribution with smaller range has less dispersion.

Quartile Deviation:- It is absolute measure of dispersion.

Define as the average amount by which two quartiles differ from median.

$$\left[\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2} \right] \quad \begin{array}{l} Q_3 - \text{Median} - \text{①} \\ Q_1 + \text{Median} - \text{②} \\ Q.D = \frac{\text{①} - \text{②}}{2} \end{array}$$

is it is the average of difference between the third and first quartile.

- * When Q.D is small, it describes small variation of the central 50% item and high Q.D means that variation among the central items is large.

Coefficient of Q.D:- It is the

relative measure of dispersion, is calculated as

$$\text{coefficient of Q.D} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

It is used to compare the degree of variation in different distribution.

Q1 Calculate the value of quartile deviation and its coefficient from the following data

Roll No:-	1	2	3	4	5	6	7
Marks:	20	28	40	12	30	15	50

Soln:- Marks in ascending order
12, 15, 20, 28, 30, 40, 50

$$Q_1 = \text{Size of } \left(\frac{N+1}{4}\right)^{\text{th}} \text{ item}$$

$$Q_1 = \text{Size of } \left(\frac{7+1}{4}\right)^{\text{th}} = 2^{\text{nd}} \text{ item}$$

$$Q_1 = 15$$

$$Q_3 = \text{Size of } \frac{3(N+1)}{4}^{\text{th}} \text{ item}$$

$$= \text{Size of } 6^{\text{th}} \text{ item}$$

$$Q_3 = 40$$

$$Q.D = \frac{Q_3 - Q_1}{2} = \frac{40 - 15}{2}$$

$$Q.D = \frac{25}{2}$$

$$Q.D = 12.5$$

$$\text{Coefficient of } Q.D = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{40 - 15}{40 + 15}$$

$$Q.D = 0.455$$

Q2 Compute Q.D and its coefficient from the following data:-

Marks	10	20	30	40	50	60
No. of Students	4	7	15	8	7	2

<u>Soln</u>	Marks	freq	cf.
	10	4	4
	20	7	11
	30	15	26
	40	8	34
	50	7	41
	60	2	43

$$N = 43$$

$$Q_1 = \text{Size of } \left(\frac{N+1}{4} \right)^{\text{th}} \text{ item}$$

$$= \text{Size of } \left(\frac{43+1}{4} \right)^{\text{th}} \text{ item}$$

$$Q_1 = \text{Size of } 11^{\text{th}} \text{ item}$$

$$\boxed{Q_1 = 20}$$

$$Q_3 = \text{Size of } \frac{3(N+1)}{4}^{\text{th}} \text{ item}$$

$$Q_3 = \text{Size of } 33^{\text{th}} \text{ item}$$

$$\boxed{Q_3 = 40}$$

$$Q.D = \frac{Q_3 - Q_1}{2} = \frac{40 - 20}{2}$$

$$\boxed{Q.D = 10}$$

$$\text{coeff. of } Q.D = \frac{Q_3 - Q_1}{Q_3 + Q_1} = 0.333$$

Q. compute the value of Q.D and its coeff. from the following data.

class:	10-20	20-30	30-40	40-50	50-60	60-70	70-80
freq:	12	19	5	10	9	6	6

Soln.	class	freq.	c.f
	10-20	12	12
	20-30	19	31
	30-40	5	36
	40-50	10	46
	50-60	9	55
	60-70	6	61
	70-80	6	67
	N = 67		

$Q_1 = \text{Size of } \frac{N}{4}^{\text{th}} \text{ item}$

$= \text{Size of } \frac{67}{4}^{\text{th}} \text{ item}$

$Q_1 = \text{Size of } 16.75^{\text{th}} \text{ item}$

ie Q_1 lies in the class 20-30.

$Q_3 = \text{Size of } \frac{3N}{4}^{\text{th}} \text{ item}$

$= \text{Size of } 50.25^{\text{th}} \text{ item}$

Q_3 lies in the class 50-60

$$\text{ie } Q_1 = l + \frac{h}{f} \left(\frac{N}{4} - c \right)$$

$$l = 20, h = 10, f = 19$$

$$c = 12$$

$$Q_1 = 20 + \frac{10}{19} (16.75 - 12)$$

$$Q_1 = 20 + 2.5$$

$$Q_1 = 22.5$$

$$Q_3 = l + \frac{h}{f} \left(\frac{3N}{4} - c \right)$$

$$l = 50, \quad h = 10, \quad f = 9, \quad c = 46$$

$$Q_3 = 50 + \frac{10}{9} (50.25 - 46)$$

$$Q_3 = 54.72$$

$$Q.D = \frac{Q_3 - Q_1}{2}$$

$$= \frac{54.72 - 22.5}{2}$$

$$Q.D = 16.11$$

$$\text{coeff. of } Q.D = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

$$\text{coeff. of } Q.D = 0.4172$$

The method of dispersion discussed above, namely, range and quartile deviation, are not measures of dispersion in strict sense as they do not show the scatterness around the average.

To study the formation of distribution we should take the deviation from the average. The other two measures, namely average deviation and the standard deviation, help us in achieving this goal.

Mean Deviation

- 1) Mean Deviation also called the average deviation
- 2) It measures the scatter of the individual observations around a central value usually the mean or median of ~~of~~ distribution.
- 3) It is defined as the arithmetic mean of differences of each observation from mean or median of distribution.
- (4) There is an advantage of taking the deviation from median because sum of ^{absolute} deviations of items from median is minimum, when signs are ignored.
- (5) If the deviations are taken from the mean, then it is called mean deviation about mean. On the other hand if deviations are taken from the median, it is called mean deviation about the median.

Mean Deviation About Mean:-

Let $\bar{x}$ be the mean of data

$$1) \left[M.D = \frac{\sum_{i=1}^N |X_i - \bar{x}|}{N} \right] \quad \text{(For individual observations)}$$

$$2) \left[M.D = \frac{\sum_{i=1}^N f_i (X_i - \bar{x})}{N} \right] \quad \text{(For Discrete series or grouped data)}$$

$$3) \left[M.D = \frac{\sum_{i=1}^N f_i (m_i - \bar{x})}{N} \right] \quad \text{(For continuous series)} \\ \text{where } m_i \text{ is mid point of classes.}$$

Mean Deviation about the Mean:-

Same formulas as above, but deviations are taken from the median of data.

Let $\hat{x}$ be the median

ie For individual observations

$$\text{Mean deviation} = \frac{\sum_{i=1}^N |x_i - \hat{x}|}{N}$$

$$\text{or} = \frac{\sum |D|}{N}$$

For grouped data

$$\text{Mean deviation} = \frac{\sum_{i=1}^N f_i |x_i - \hat{x}|}{N}$$

$$= \frac{\sum f |D|}{N}$$

Coefficient of Mean Deviation:-

It is the relative measure corresponding to mean deviation.

$$\text{coefficient of M.D} = \frac{\text{M.D}}{\text{Mean}}$$

average about which deviations are taken.

$$\text{ie coeff. of M.D} = \frac{\text{M.D}}{\text{Mean}} \text{ or } \frac{\text{M.D}}{\text{Median}}$$

Q1 calculate the mean and its coefficient of following two group of incomes.

Group I: 4000, 4200, 4400, 4600, 4800

Group II: 3000, 4000, 4200, 4400, 4600, 4800, 5800

For Group I :-

$$\text{Mean} = \frac{\sum X}{N} = \frac{\sum X}{5} = 4400$$

$$\text{Median} = \text{Size of } \left(\frac{5+1}{2}\right)^{\text{th}} \text{ item}$$

$$\text{Median} = 4400$$

(Mean and Median are equal in this case)

X	Deviation from Mean D
4000	400
4200	200
4400	0
4600	200
4800	400
N=5	$\sum D = 1200$

$$\begin{aligned} \text{Mean Deviation} &= \frac{\sum |x_i - \bar{x}|}{N} = \frac{\sum |D|}{N} \\ &= \frac{1200}{5} = 240 \end{aligned}$$

For Group IInd :-

$$\text{Mean} = 4400 = \bar{x}$$

$$\text{Median} = 4400$$

X	Deviation from mean D = $ x_i - \bar{x} $
3000	1400
4000	400
4200	200
4400	0
4600	200
4800	400
5800	1400
N=7	$\sum D = 4000$

Mean Deviation

$$= \frac{\sum |x_i - \bar{x}|}{N} = \frac{\sum |D|}{N}$$

$$= \frac{4000}{7} = 571.43$$

coeff of M.D for group I :-

$$= \frac{M.D}{\text{Mean}} = \frac{240}{4400}$$

$$\text{coeff of M.D} = 0.0545$$

coeff of M.D for group II :-

$$= \frac{M.D}{\text{Mean}} = \frac{571.43}{4400} = 0.1299$$

Q2 calculate Mean deviation and its coefficient

X :	10	11	12	13	14
f :	3	12	18	12	3

Soln we will find mean deviation from Median.

X	f	cf	$ D = x_i - \hat{x} $	f D
10	3	3	2	6
11	12	15	1	12
12	18	33	0	0
13	12	45	1	12
14	3	48	2	6
N = 48			$\Sigma D = 36$	$\Sigma f D = 36$

Median = Size of $\frac{N}{2}$ item

= Size of 24th item

ie Median = 12

$$\text{Mean deviation} = \frac{\Sigma f |D|}{N} = \frac{36}{48} = 0.75$$

Q3 Calculate the mean deviation from the following data:-

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No of Stud:	4	6	10	20	10	6	4

Soln

Marks	f	cf	m _#	m - 35 = D	f D
0-10	4	4	5	30	120
10-20	6	10	15	20	120
20-30	10	20	25	10	100
30-40	20	40	35	0	0
40-50	10	50	45	10	100
50-60	6	56	55	20	120
60-70	4	60	65	30	120
N = 60				$\Sigma f D = 680$	

Median class corresponds to $\frac{N}{2} = 30$ i.e.
Median class is 30-40

$$\begin{aligned}\text{Median} &= l + \frac{h}{f} \left(\frac{N}{2} - c \right) \\ &= 30 + \frac{10}{20} (30 - 20)\end{aligned}$$

$$\hat{x} = \text{Median} = 30 + 5 = 35$$

$$M.D = \frac{\Sigma f|D|}{N} = \frac{680}{60} = 11.33$$

$$\boxed{M.D \text{ from Median} = 11.33}$$

Q calculation mean deviation from median from the following data

Marks less than:	80	70	60	50	40	30	20	10
No. of students:	100	90	80	60	32	20	13	5

Hint:- arrange in ascending order of various classes.